"No Motion"

)

frame1 = frame2_gray

time.sleep(1)

7. Result:

PRACTICAL-17

1. Aim

push-button and an LED with a web dashboard. The flow must allow users to toggle the LED from the dashboard and via the physical button. A button press should force the LED ON Design and implement a Node-RED flow on Raspberry Pi that integrates a physic for 5 seconds and then automatically restore the LED to the dashboard's last state.

2. Prior Concepts

i) Concept of Python Programming

ii) LED interfacing with Raspberry Pi

iii) Push button switch operation

iv) Web browser operation

v) Event-driven programming basics

3. New Concepts

i) Node-RED visual programming.

ii) Flow-based loT application development.

iii) Web-based device control.

iv) Dashboard and hardware synchronization.

v) Timer and delay node implementation.

4. Hardware Required

i) Push Button Switch

ii) LED

iii) Jumper Wires

5. Procedure

Pin Connection

Camera Module: Connect the Raspberry Pi Camera Module to the CSI Camera Port of Raspberry Pi 4

LED Connections to Raspi:

Component	Raspberry Pi 4
LED Anode (+)	GPIO18 (Pin 12) through 220 resistor
LED Cathode (-)	GND (Pin 6)

Input IMAGE

i) Install Python3 on Raspi.

Write the below code in command prompt:

sudo apt install idle3

Install Node-RED Dashboard:

cd ~/.node-red

npm install node-red-dashboard

Restart Node-RED:

node-red-stop

node-red-start

6. Execution

Now open Node Red in browser:

http://<RaspberryPi-IP>:1880

PRACTICAL-18

1.Aim

Write a program to read the CPU%, Memory% and CPU temperature of Raspberry Pi,

2. Prior Concepts

i) Linux operating system fundamentals

ii) Processor and memory concepts.

3. New Concepts

i) System performance monitoring.

ii) CPU utilization measurement.